MENTAL HEALTH DISORDER PREDICTION

Using the DASS-42 Self-Assessment Questionnaire

A Complete Case Study Report

1. Problem Statement

Mental health disorders such as depression, anxiety, and stress-related conditions continue to be significantly underdiagnosed across populations worldwide. According to the World Health Organization, a substantial proportion of individuals experiencing psychological distress do not receive timely clinical evaluation or intervention. Delays in identification often lead to worsening symptoms, reduced quality of life, impaired productivity, and, in severe cases, increased risk of self-harm. Despite growing awareness, accessible and scalable early-screening mechanisms remain limited, particularly in non-clinical and telehealth environments.

This case study aims to address this gap by developing a data-driven predictive model capable of identifying individuals at risk of mental health disorders based on self-reported symptom data.

The project utilizes the publicly available “Depression, Anxiety, Stress Scales Responses” dataset from Kaggle, which contains approximately 40,000 responses across 42 questionnaire items derived from the standardized DASS-42 instrument, along with demographic attributes. The DASS-42 framework measures severity across three psychological dimensions depression, anxiety, and stress through structured symptom-based questions scored on a four-point scale.

1.1 Dataset References

"Depression, Anxiety, Stress Scales Responses" dataset from Kaggle. It contains 42 features (DASS-42) questionnaire items on depression/anxiety/stress symptoms, demographics.

Official Document for DASS-42 questionnaire and Score –

Link 1 - https://upcommons.upc.edu/server/api/core/bitstreams/c357c588-e151-41c0-991e-45dd2f3303a5/content#:~:text=Depression:%203%2C%205%2C%2010,S%20columns%20(Table%201).

Link 2 - https://www.scribd.com/document/523872913DASS-42-Scoring#:~:text=The%20DASS%2D42%20uses%20a%204%2Dpoint%20scale%20ranging,and%20extremely%20severe%20levels%20for%20each%20state.

1.2 The Three Prediction Tasks

Rather than lumping everything into one model, we built three separate models — one for each condition. This makes sense because each condition has its own dedicated set of questions, different severity thresholds, and different clinical significance:

Condition	Relevant Questions	Test Set Size	Class Distribution (before SMOTE)
Depression	14 questions (Q3,Q5,Q10,Q13,Q16,Q17,Q21,Q24,Q26,Q31,Q34,Q37,Q38,Q42)	11,930 rows	Extremely Severe: 13,572 | Normal: 8,853 | Moderate:

			7,077 | Severe: 6,477 | Mild: 3,785
Anxiety	14 questions (Q2,Q4,Q7,Q9,Q15,Q19,Q20,Q23,Q25,Q28,Q30,Q36,Q40,Q41)	11,930 rows	Extremely Severe: 14,118 | Normal: 9,723 | Moderate: 7,047 | Severe: 6,112 | Mild: 2,764
Stress	14 questions (Q1,Q6,Q8,Q11,Q12,Q14,Q18,Q22,Q27,Q29,Q32,Q33,Q35,Q39)	11,930 rows	Normal: 11,794 | Moderate: 8,730 | Severe: 8,572 | Extremely Severe: 5,747 | Mild: 4,921

1.3 The Five Severity Levels

Each condition (Depression, Anxiety, Stress) is independently classified into one of five levels:

Level	Label	What It Means in Practice
0	Normal	Scores fall within the healthy expected range
1	Mild	Slightly elevated — worth monitoring, not yet clinical
2	Moderate	Noticeably elevated — professional guidance is advisable
3	Severe	Significantly impaired — professional intervention needed
4	Extremely Severe	Highest severity — urgent professional care required

2. Data Overview

The dataset comes from Kaggle and contains responses from real people who voluntarily completed the DASS-42 questionnaire online. The raw file had 39,775 rows (one per respondent) and 172 columns.

Column Group	Count	Examples
Question Answers (QxA)	42	Q1A, Q2A ... Q42A
Time Per Question (QxE)	42	Q1E, Q2E ... Q42E — milliseconds spent
Question Order (QxI)	42	Q1I ... Q42I — the order shown to user
Demographics	11+	age, gender, education, urban, religion, race, orientation, married, familysize, screensize, engnat, country, major
Other	35+	TIPI personality scores, VCL vocabulary scores

Every question answer (Q1A through Q42A) records the respondent's answer on a 1 to 4 scale in the raw file. I converted this to 0 to 3 by subtracting 1 from every answer — this matches the official DASS-42 scoring guide, where 0 means 'did not apply at all' and 3 means 'applied very much or most of the time.

2.1 Severity Scores Calculation logic

For each respondent, I summed their answers to the 14 condition-specific questions. That total becomes their raw score for that condition. The score is then mapped to a severity level using official DASS-42 clinical cutoffs:

Severity Level	Depression Score Range	Anxiety Score Range	Stress Score Range
Normal (0)	0 – 9	0 – 7	0 – 14
Mild (1)	10 – 13	8 – 9	15 – 18
Moderate (2)	14 – 20	10 – 14	19 – 25
Severe (3)	21 – 27	15 – 19	26 – 33
Extremely Severe (4)	28 and above	20 and above	34 and above

2.2 The At-Risk Label

In addition to the three multiclass predictions, we also created a binary at-risk label. A person was flagged as 'At Risk' if they scored Moderate or above (level 2+) on ANY of the three conditions. The numbers were revealing:

Insight: Over 80% of the 40,000 respondents showed at least moderate symptoms of depression, anxiety, or stress. It it reflects that online DASS surveys tend to attract people who are already struggling. Even so, the finding underscores how widespread these conditions are and why early screening matters.

3. Exploratory Data Analysis (EDA)

Before building any model, I spent time looking at the data carefully — visualising distributions, comparing groups, and looking for patterns. This step is essential because it shapes every decision that follows. Here is everything I found.

3.1 Score Distributions - Where do people land?

I plotted histograms of the three raw scores across the entire dataset. The results painted a clear picture:

- **Depression:** The distribution is NOT bell-shaped — it peaks in the Mild-to-Moderate range and has a long right tail. This means many people score moderately high, while a significant portion scores at the Extremely Severe end. Very few people score near zero (Normal).
- **Anxiety:** Anxiety scores are right-skewed — a large number of people cluster at zero or near-zero, but there is a sharp secondary peak at the Extremely Severe end. This bimodal pattern suggests the population splits into 'not very anxious' and 'very anxious' groups with fewer people in between.
- **Stress:** Stress follows a broader, more spread-out distribution centred around the Moderate range (scores 15–25). More people fall into the Normal category for stress than for depression or anxiety.

Insight: Extremely Severe is the Most Common Category for Depression and Anxiety

Across depression and anxiety, the 'Extremely Severe' class was actually the largest single group in the dataset. For depression, 13,572 respondents (34%) scored Extremely Severe. For anxiety, 14,118 (36%) were Extremely Severe. This is a critical finding — it confirms the dataset is not representative of the general population but rather of people actively experiencing mental health challenges.

3.2 Age and Risk

I compared age distributions between At Risk and Not At Risk groups using histograms and boxplots. The statistical breakdown was:

Group	Count	Mean Age	Median Age	Std Dev	Age Range
Not At Risk	7,830	26.31	23	10.40	13 – 79
At Risk	31,934	22.68	21	7.88	13 – 80

Insight 2: Young Adults (18–25) Are at Highest Risk

Participants aged 18 to 25 dominated the high-risk category. This age bracket coincides with university, early career transitions, financial independence, and relationship changes all well-

documented stress multipliers. The finding has practical implications: mental health screening in educational institutions and early-career workplaces could have substantial impact.

3.3 Gender and Risk

We analysed the at-risk rate broken down by gender (coded as 1 = Male, 2 = Female, 3 = Other in the dataset):

Gender	At Risk Rate	Key Observation
Male	~75–78%	High but slightly lower than female
Female	~80–83%	Slightly higher risk rate than male
Other / Non-Binary	>90%	Highest at-risk rate of all three groups

Insight 3: Gender Patterns Are Significant — and Require Careful Handling

Two things stand out. First, the 'Other' gender category shows over 90% at-risk rates — suggesting that people outside the gender binary face disproportionately high mental health burden, which aligns with broader research on LGBTQ+ mental health. This is flagged as a fairness concern to address before deployment.

3.4 Education and Risk

Risk rates varied significantly by education level. Lower education correlated with higher at-risk rates:

Education Level	At-Risk Rate (Approx.)	Pattern
Less than High School	Highest	Least economic and social resources
High School	High	Transition period, limited support
College	Moderate	Mix of stress and growing resources
Bachelor's Degree	Moderate-Low	More support, higher awareness
Graduate Degree	Lowest	Most resources, highest health literacy

Insight: Education as a Protective Factor

Higher education appears to be protective — not necessarily because education itself reduces mental health risk. This finding suggests that screening tools and interventions should be especially targeted at lower-education populations who may have less access to support.

3.5 Severity Distribution Across Conditions

When we looked at the full breakdown of severity levels across all three conditions, a clear pattern emerged:

Condition	Normal	Mild	Moderate	Severe	Extremely Severe
Depression	8,853 (22%)	3,785 (10%)	7,077 (18%)	6,477 (16%)	13,572 (34%)
Anxiety	9,723 (24%)	2,764 (7%)	7,047 (18%)	6,112 (15%)	14,118 (36%)
Stress	11,794 (30%)	4,921 (12%)	8,730 (22%)	8,572 (22%)	5,747 (14%)

Insight 5: Stress Has a More Even Distribution — Depression and Anxiety Are Dominated by Extremes

Stress shows a more balanced spread across severity levels, with Normal being the largest single class (30%). Depression and Anxiety tell a very different story — they are dominated by the Extremely Severe class. The 'Mild' class is the smallest in all three conditions, making it the hardest class to train on and the most likely to be misclassified. This directly impacts model performance on that class.

3.6 Correlation Between Conditions

We computed correlations between the three score columns to understand how Depression, Anxiety, and Stress relate to each other:

Pair	Correlation Coefficient	Interpretation
Depression vs Anxiety	~0.72	Strong positive — they co-occur frequently
Depression vs Stress	~0.68	Strong positive — often appear together
Anxiety vs Stress	~0.65	Strong positive — linked but slightly less so
Any Score vs Age	~-0.10	Weak negative — younger respondents score slightly higher

Insight: These Conditions Rarely Come Alone

The high correlations mean someone who scores Severely on depression is very likely to also show elevated anxiety or stress. This is consistent with clinical understanding — comorbidity is the norm, not the exception. For a screening tool, this means identifying risk in one dimension should prompt checking all three.

4. Preprocessing and Feature Engineering

4.1 Selecting the Right Features

For each model, I selected:

Condition questions: 14 DASS question answer columns specific to that condition (e.g., Q3A, Q5A, Q10A, Q13A, Q16A, Q17A, Q21A, Q24A, Q26A, Q31A, Q34A, Q37A, Q38A, Q42A for Depression)

Demographics: 11 demographic columns: education, urban, gender, engnat (English as native), age, screensize, religion, orientation, race, married, familysize.

This gave 25 features per model (14 question answers + 11 demographics). Time-per-question and question-order columns were excluded — they do not tell us about the person's mental state, only how they interacted with the survey interface.

4.2 Age Binning

The raw age column was converted into six age brackets rather than using the raw number. This is standard practice because age does not have a purely linear relationship with mental health outcomes — the difference between 18 and 24 is clinically different from the difference between 40 and 46, even though the numerical gap is the same.

4.3 Train-Test Split

The data was split 70% for training and 30% for testing. The split was stratified — meaning each severity class kept the same proportion in both train and test sets. This prevents a lucky or unlucky split from distorting the evaluation numbers.

4.4 Handling Class Imbalance with SMOTE

The biggest challenge in this dataset was that the severity classes were not evenly represented. If we trained a model on unbalanced data, it would learn to favour the majority class ('Extremely Severe' for depression) and largely ignore the minority class ('Mild').

After SMOTE, training set sizes became:

Condition	Before SMOTE (train)	After SMOTE (train)	Balanced Class Size
Depression	27,834	47,500	9,500 per class
Anxiety	27,834	49,410	9,882 per class
Stress	27,834	41,275	8,255 per class

4.5 Feature Scaling

All features were scaled using StandardScaler — this centres every feature around a mean of 0 with a standard deviation of 1. Without scaling, features with large numerical ranges (like age, which goes up to 80) would dominate features with small ranges (like DASS answers, which only go 0–3). Scaling ensures every feature competes on equal footing.

5. Model Training

We trained five different types of classification models for each of the three conditions, giving us fifteen models in total. The goal was systematic comparison — not guessing which model would work best, but letting the data decide.

5.1 First Attempt: Binary Classification

Before building the multiclass models, I first tried a simpler binary classification:

Binary Classification Result: 96.4% Accuracy

Logistic Regression on the binary task achieved 96.4% accuracy — impressive on paper, but misleading. The dataset is 80% At Risk, so a naive model that always predicts 'At Risk' would get 80% accuracy without learning anything. The real test is the per-class performance: 91% F1 for Not At Risk and 98% F1 for At Risk. The high binary accuracy led us to pursue the harder multiclass problem.

5.2 Depression Model Results

The Depression dataset had 39,764 rows with 25 features each. The most common class was Extremely Severe (34%) and the rarest was Mild (10%). After SMOTE balancing, all five classes had 9,500 training examples. Here are the results of all five models on the 11,930-row test set:

All Five Model Scores — Depression

Model	Accuracy	F1-Score	Precision	Recall	Verdict
Logistic Regression	0.881	0.883	0.896	0.881	BEST — Winner
Decision Tree	0.660	0.674	0.718	0.660	Too simple — rejected
KNN	0.846	0.850	0.863	0.846	Good but second to LR
Random Forest	0.827	0.832	0.854	0.827	Solid but not best
XGBoost	0.850	0.853	0.868	0.850	Strong but LR edges it out

5.3 Anxiety Model Results

The Anxiety dataset had a particularly challenging class: Mild had only 2,764 examples (7% of total) — by far the smallest class. After SMOTE, each class had 9,882 training examples:

Model	Accuracy	F1-Score	Precision	Recall	Verdict
Logistic Regression	0.823	0.830	0.860	0.823	BEST — Winner

Decision Tree	0.599	0.623	0.682	0.599	Worst performer
KNN	0.780	0.792	0.822	0.780	Decent but third place
Random Forest	0.745	0.756	0.802	0.745	Surprisingly weak for ensemble
XGBoost	0.778	0.787	0.820	0.778	Second place but behind LR

Insight: Anxiety's Mild Class is the Worst in the Entire Study

The Mild anxiety class has an F1 of just 0.31 — the weakest single-class performance in the entire project. With only 829 test examples (7% of anxiety test data), even after SMOTE this class is genuinely hard. The Mild-Normal and Mild-Moderate boundaries are clinically fuzzy. This is where clinical review should always accompany the model prediction.

5.4 Stress Model Results

Stress had the most balanced class distribution of the three conditions — Normal was largest at 30% and Mild was smallest at 12%. This balance made training relatively easier:

Model	Accuracy	F1-Score	Precision	Recall	Verdict
Logistic Regression	0.871	0.873	0.884	0.871	BEST — Winner
Decision Tree	0.603	0.618	0.662	0.603	Poor — same issue as other conditions
KNN	0.776	0.783	0.816	0.776	Third place
Random Forest	0.813	0.818	0.835	0.813	Good but not best
XGBoost	0.838	0.841	0.852	0.838	Strong second place

6. Model Checks – Best Model

6.1 Overfitting Check Results — Best Model (Logistic Regression)

Condition	Training Accuracy	Test Accuracy	Verdict
Depression	0.878 (87.8%)	0.881 (88.1%)	No overfitting — slightly better on unseen data

Anxiety	0.801 (80.1%)	0.823 (82.3%)	No overfitting — generalises well to new data
Stress	0.849 (84.9%)	0.871 (87.1%)	No overfitting — performs better on new data

6.2. Cross-Validation — Is the Model Stable Across Different Data Splits?

A single train-test split can be lucky or unlucky. Cross-validation (CV) repeats the evaluation 5 times, each time using a different 20% of the training data as a validation set.. The standard deviation across the 5 folds tells us how stable the model is.

Condition	Mean CV Score	Std Dev	Stability Verdict
Depression (Best LR)	0.853	0.026	Stable — low variance across folds
Anxiety (Best LR)	0.760	0.072	High variance — sensitive to data distribution
Stress (Best LR)	0.820	0.053	Moderate variance — one weak fold (0.716)

Insight: Anxiety Has High Variance — It Is Data-Sensitive

The anxiety model shows a CV standard deviation of 0.072 — nearly three times that of the depression model. The weakest fold scored only 0.618. This means the anxiety model's performance depends heavily on which subset of data it trains on. The underlying cause is the tiny Mild class (7% of anxiety data) — when a fold happens to have fewer Mild examples, performance drops. This was flagged for treatment during hyperparameter tuning via stronger regularisation.

6.3. Multicollinearity check (Top Features)

Condition	Highest VIF Feature	VIF	Second Highest	VIF
Depression	Q21A (difficulty rousing)	3.66	Q38A (not able to feel positive)	3.53
Anxiety	Q10A (feeling dizzy)	2.29	Q31A (feeling of dread)	2.14
Stress	Q13A (easily agitated)	2.53	Q3A (over-react to situations)	2.38

Insight: Multicollinearity Is Acceptable — No Serious Concern

The highest VIF value found across all three conditions is 3.66 - comfortably below the 5.0 threshold that would indicate moderate concern and well below the 10.0 threshold that would

require action. Some correlation among DASS questions is expected (symptoms of the same condition naturally cluster), but it is not severe enough to harm model reliability. No features need to be removed.

6.4. Hyperparameter Tuning

I used GridSearchCV to systematically try multiple combinations of hyperparameter values and evaluate each using 5-fold cross-validation. For Logistic Regression, we searched over combinations of C values, solver algorithms, and penalty types.

Tuning produced no improvement in CV score for any of the three conditions. This is not a failure — it is confirmation that the initial hyperparameters chosen for Logistic Regression were already close to optimal for this dataset. The anxiety model's variance (CV std=0.072) is driven by the data distribution, not the hyperparameters. Further improvement would require either more Mild-class training examples or a different modelling approach for anxiety.

7. Final Evaluation on Unseen Test Data

After tuning, we ran each best model one final time on the completely held-out test set — data that was never used during training, SMOTE balancing, scaling, or tuning. These are the most honest numbers in the entire project:

Model	Accuracy	F1-Macro	Recall	Precision	Overall Verdict
Depression (Logistic Regression)	0.882 (88.2%)	0.838	0.849	0.840	Excellent — best of the three
Anxiety (Logistic Regression)	0.823 (82.3%)	0.735	0.750	0.745	Good — held back by Mild class
Stress (Logistic Regression)	0.872 (87.2%)	0.862	0.876	0.855	Excellent — most balanced overall

The Most Important Fairness Finding

The notebook's own conclusion is clear: 'The model shows significant fairness gaps across age, gender, marital status, and family size, indicating demographic imbalance and potential bias that must be addressed before deployment.' This is not a minor caveat — it is a fundamental requirement. Any deployment without addressing these gaps would disproportionately harm the demographic groups the tool performs worst on.

8. Model Deployment & API

7.1 API Architecture

The saved models power a FastAPI web service (app_with_drift.py). The API accepts a POST request with 42 DASS answers plus demographic data, validates all inputs using Pydantic schemas (ensuring all answers are in the 0–3 range), extracts the correct question subsets for each condition, scales the features, runs all three predictions, and returns severity levels for Depression, Anxiety, and Stress — plus a drift detection flag.

7.2 Data Drift Monitoring

The deployment includes automatic drift monitoring using two statistical tests: PSI (Population Stability Index) and the Kolmogorov-Smirnov test. These compare incoming real-world prediction requests against the reference distribution from training data. If the incoming data starts looking significantly different from the training distribution — for example, a sudden shift in the age profile of users — the API flags 'drift_detected: true' and logs a warning. This tells the team: the world has changed and the model may need retraining.

The production API incorporates a Drift Monitor module that detects when incoming data distributions diverge from the training distribution. Two statistical tests are combined:

• PSI (Population Stability Index) – flags drift when PSI > 0.25

• KS-Test (Kolmogorov-Smirnov) – flags drift when p-value < 0.05

• Both tests are run on scaled feature vectors for each of the three subscales

• All drift events are logged to drift_logs.log with timestamps and PSI values

Reference statistics (training distribution) are persisted in reference_stats.pkl. Each prediction call computes PSI and performs a KS-test to determine if the incoming data has drifted from the training distribution.

9. Conclusions & Recommendations

Key Insights Found

- **Insight 1:** Over 80% of online DASS-42 respondents show at least moderate symptoms on one condition — confirming widespread underdiagnosis.
- **Insight 2:** Extremely Severe is the most common single class for Depression (34%) and Anxiety (36%) — the dataset reflects people who are actively struggling.
- **Insight 3:** Young adults aged 18–25 are at highest risk — mean age of At Risk group is 22.68 versus 26.31 for Not At Risk.
- **Insight 4:** All genders show >75% at-risk rates, with the 'Other' gender group exceeding 90% — supporting evidence on LGBTQ+ mental health burden.
- **Insight 5:** Lower education levels correlate with higher at-risk rates — education appears to be a protective factor.
- **Insight 6:** Depression, Anxiety, and Stress scores are strongly intercorrelated ($r \approx 0.65–0.72$) — these conditions frequently co-occur.

- **Insight 7:** 'Mild' severity is consistently the hardest class to predict across all three conditions — reflecting genuine clinical ambiguity at the Normal/Mild boundary.
- **Insight 8:** Anxiety's Mild class has F1 = 0.31 — the weakest single-class result in the study, driven by only 829 test examples (7% of anxiety data).
- **Insight 9:** Stress is the most balanced performing condition — better Mild class detection (F1=0.70) thanks to more balanced class distribution.
- **Insight 10:** No overfitting detected — test accuracy slightly exceeds training accuracy in all three conditions due to SMOTE augmenting training difficulty.
- **Insight 11:** Anxiety has high CV variance (std=0.072) — performance is sensitive to data distribution, especially the tiny Mild class.
- **Insight 12:** Multicollinearity is acceptable — highest VIF is 3.66 (Q21A in Depression), well below the 5.0 concern threshold.
- **Insight 13:** Hyperparameter tuning produced no CV improvement — the initial configurations were already near-optimal.
- **Insight 14:** SHAP analysis confirmed the models learned genuine clinical patterns: depression→anhedonia/self-worth, anxiety→physical symptoms, stress→emotional reactivity.
- **Insight 15:** Significant fairness gaps found — gender gap up to 0.216 accuracy points in anxiety, large age gaps throughout. These must be resolved before clinical deployment.

Recommendations

- Address fairness gaps: Collect more representative data from underrepresented age, gender, and racial groups. Apply fairness constraints in retraining.
- Improve Anxiety's Mild class: More training examples specifically for the Mild anxiety class, or a targeted ensemble approach for this subgroup.
- Clinical validation partnership: Pair predictions against actual clinician diagnoses on a real cohort to validate clinical accuracy beyond questionnaire-derived labels.
- Threshold calibration: For safety-critical use, tune the decision threshold to prioritise recall (catching more true high-severity cases) over precision.
- Consent and privacy infrastructure: Implement explicit informed consent, end-to-end encryption, and right-to-deletion support before any user data is collected.